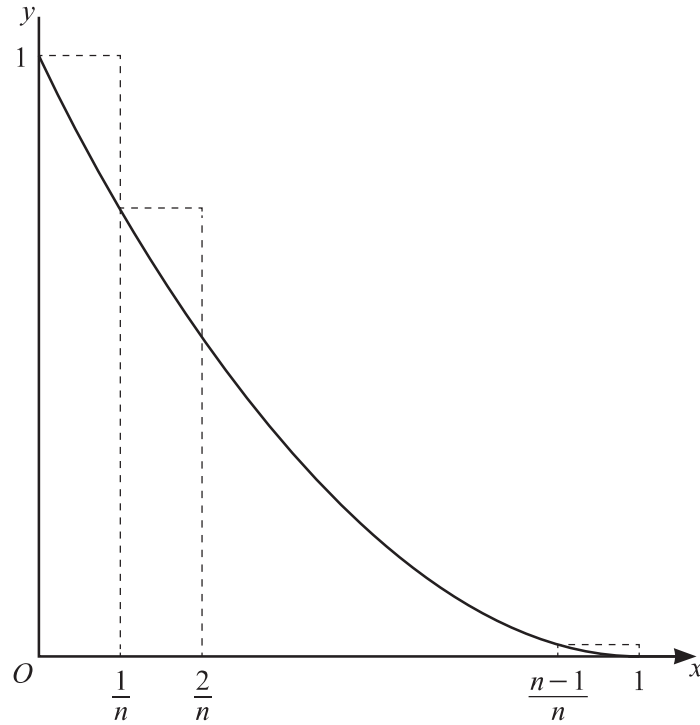


- 1 (a)** Find the Maclaurin series for $\sin^{-1}x$ up to and including the term in x^3 . [5]

[illegible]

- (b) Deduce an approximation to $\int_0^{\frac{1}{5}} \frac{1}{\sqrt{1-u^2}} du$, giving your answer as a fraction. [1]

[illegible]



The diagram shows the curve with equation $y = (1-x)^2$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

- (a) By considering the sum of the areas of these rectangles, show that $\int_0^1 (1-x)^2 dx < U_n$, where

$$U_n = \frac{2n^2 + 3n + 1}{6n^2}. \quad [5]$$

[illegible]

- (b)** Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 (1-x)^2 dx$. [4]

[illegible]

- (c) Show that $\lim_{n \rightarrow \infty} (U_n - L_n) = 0$. [2]

[illegible]

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(a) Find the exact value of I_{-1} giving your answer in the form $\ln a$, where a is an integer to be determined. [2]

[illegible]

(b) By considering $\frac{d}{dx}(x(1+x^2)^{\frac{1}{2}n})$, or otherwise, show that

$$(n+1)I_n = nI_{n-2} + \frac{4}{3}\left(\frac{5}{3}\right)^n. \quad [5]$$

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

- (c) A curve has equation $y = x^2$, for $0 \leq x \leq \frac{2}{3}$. The arc length of the curve is denoted by s .

Use the substitution $u = 2x$ to show that $s = \frac{1}{2}I_1$ and find the exact value of s . [4]

[illegible]